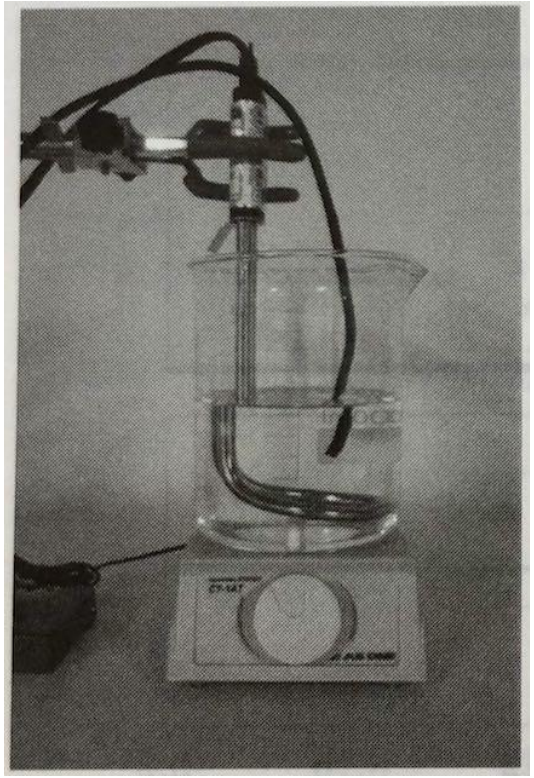
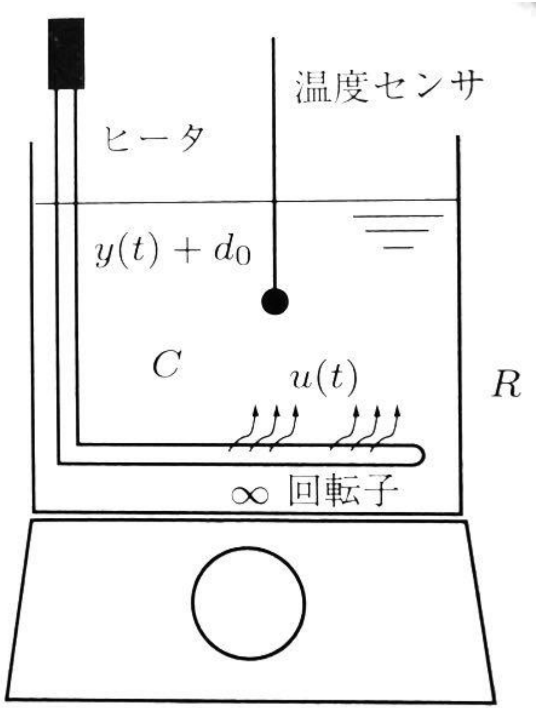
Step Response Test

Electric, Electronic and System Engineering

Water Temperature Control (1/2)

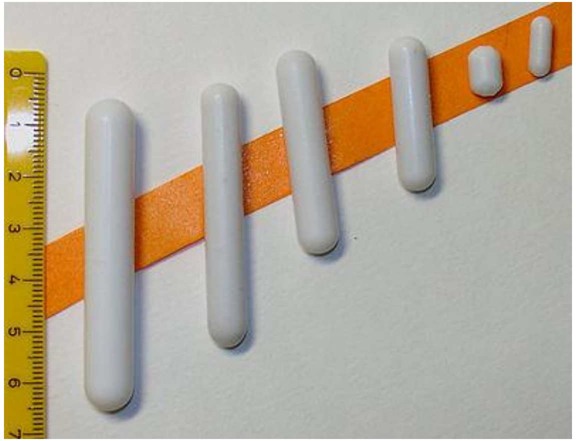
Setup





Magnetic Stirrer

a stir bar spins in a liquid
by magnetic field



Magnetic Stirrer

Inexpensive DOY Magnetic Stirrer

Equation

$$\frac{d\tilde{y}(t)}{dt} = u(t) - \frac{\tilde{y}(t) - d_0}{\quad}$$

$$\frac{dy(t)}{dt} = u(t) - \frac{y(t)}{\quad}$$

C [J/K] : Heat Capacitor

R [K/W] : Heat Resistance

d0[K] : Outside Temperature

u(t)[W] : Heat Input from Heater

y~(t) [K] : Water Temperature

y(t) [K] : Difference between Water and Outside Temperature

Laplace Transform (y(0)=0)

$$CsY(s) = U(s) - \frac{1}{R}Y(s)$$

$$Y(s) = \frac{1}{\frac{1}{R} + Cs}U(s)$$

$$= \frac{R}{1 + RCs}U(s)$$

K=R : Gain

T=RC : Response Time

$$G(s) = \frac{K}{1 + Ts}$$

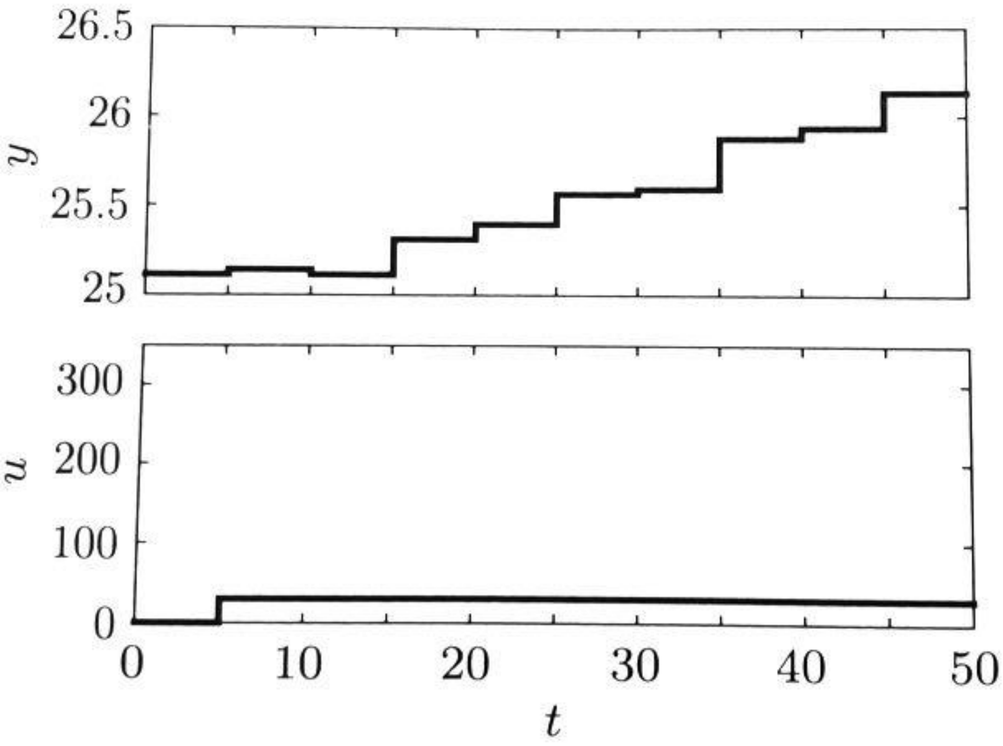
Identification of Parameters

Observe Step Input Response

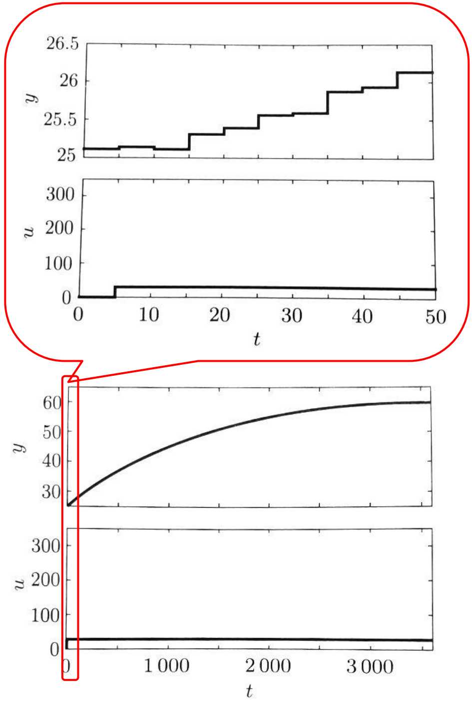
→This system has 10-15s delay.

Plant G(s) is **first order delay with dead time L**

$$G(s) = \frac{K}{1 + Ts} e^{-Ls}$$



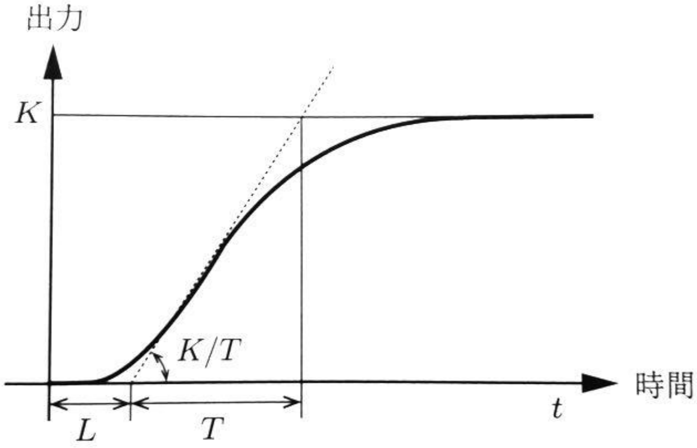
Identification from Response Curve



L

K

T



K = 1.15 K/W
L = 15 s
T = 1115 s

System Identification

Input:

- Step
- Lamp
- Sine Wave
- Maximum-length sequence

Observe:

- Time-Domain
- Frequency-Domain

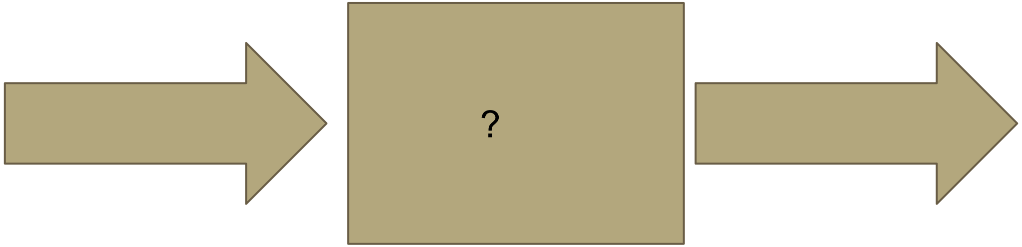


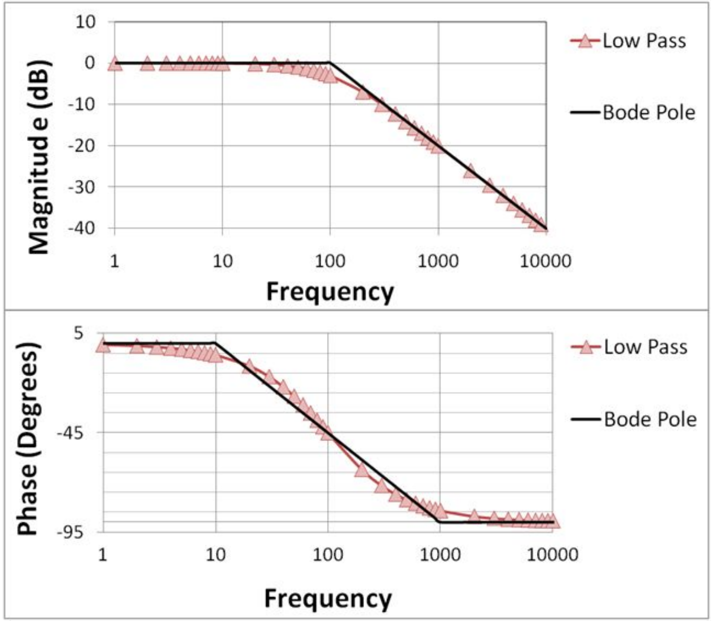
Image of System Identification

System Analysis

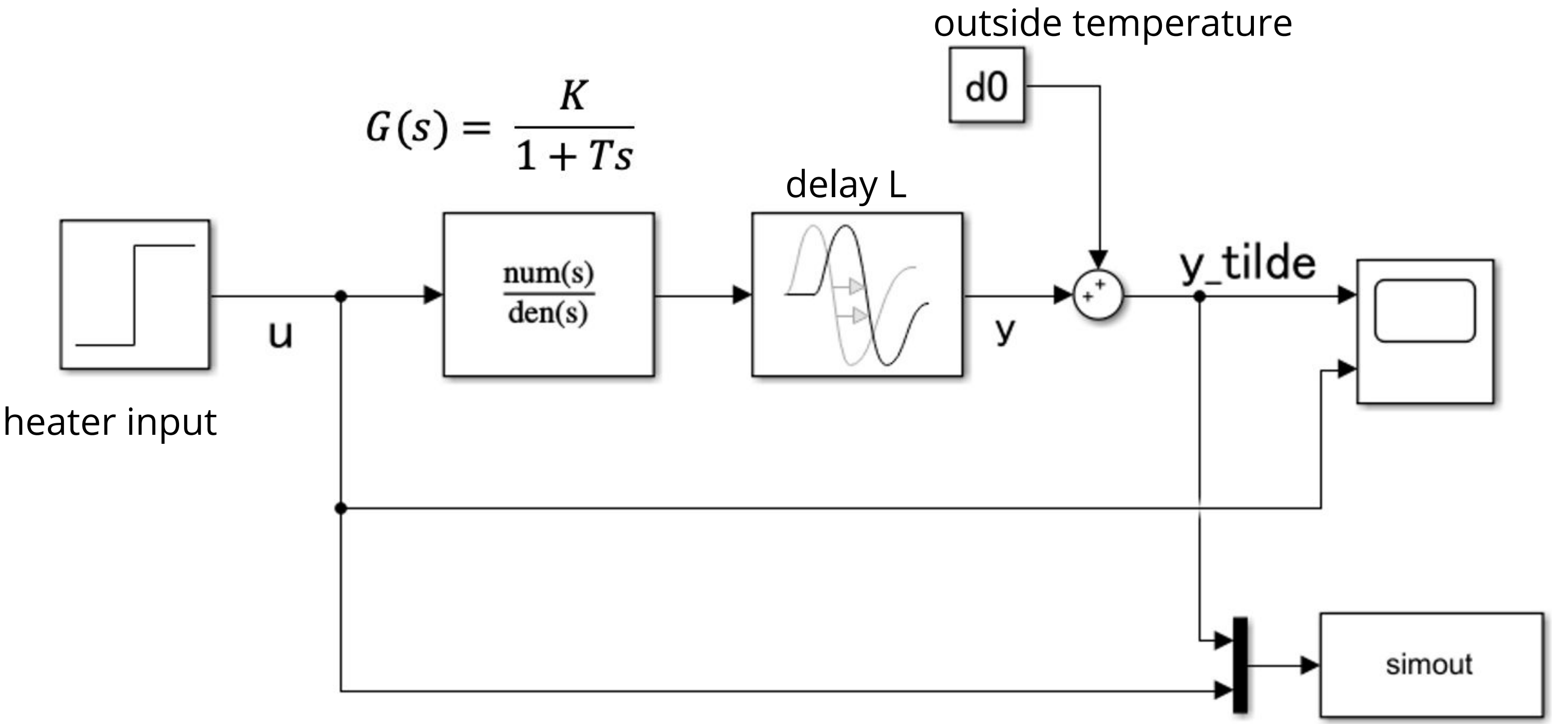
$$\frac{Y(s)}{U(s)} = \frac{K}{1+Ts}$$

?

System Identification



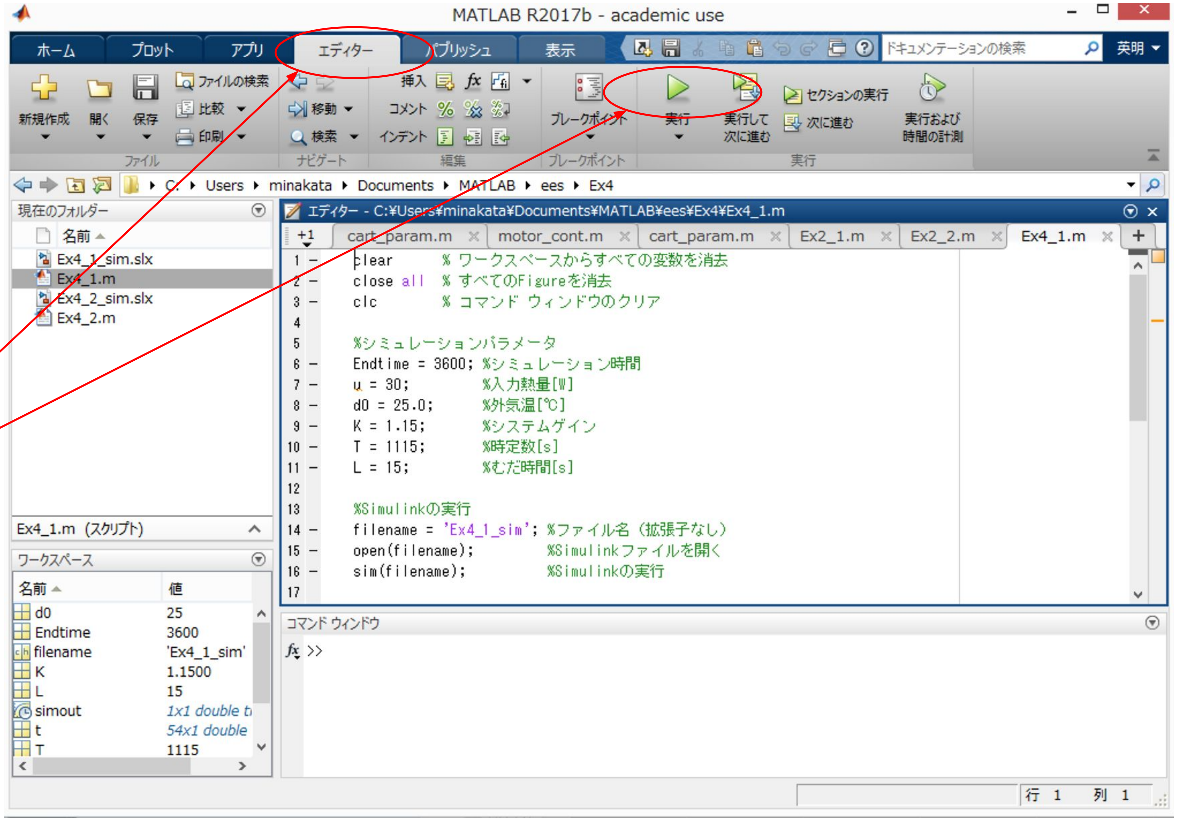
Implement Simulink Model



Control Simulations using m.file

Ex4_1.m controls simulation named Ex4_1_sim

1. open 'Ex4_1.m'
2. move to 'editor' tab
3. click 'run'

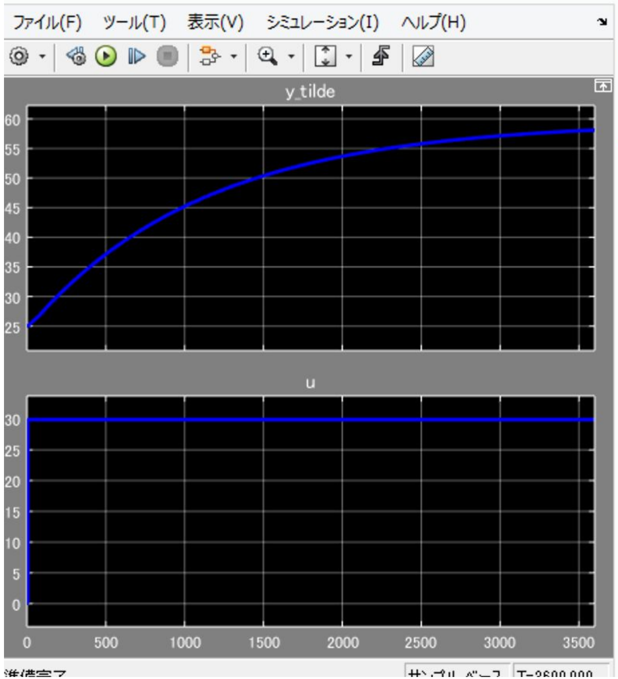


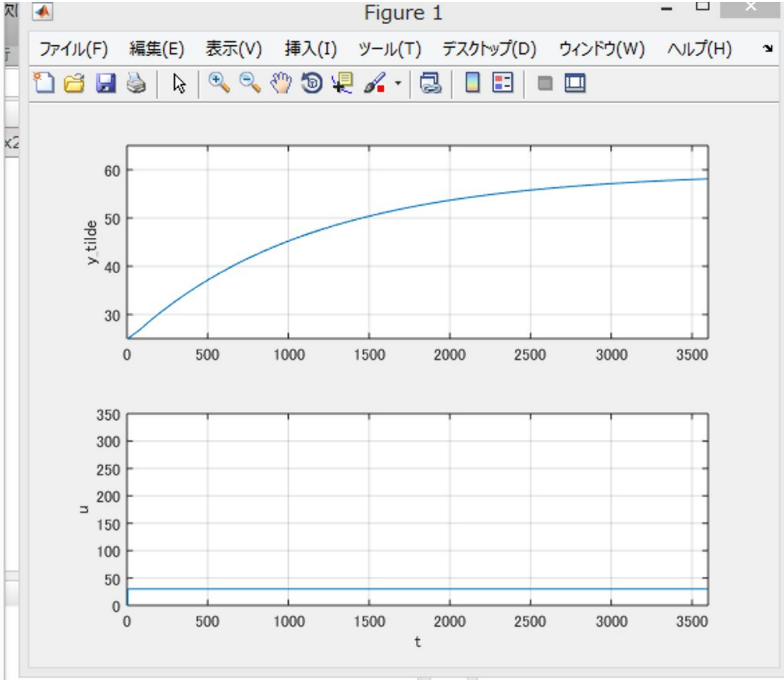
Example Output

Heater :30W input

Start :25℃

Final :59℃





Caution! Do not use large input for step test

but why?

- Does large input damage the plant ?
-

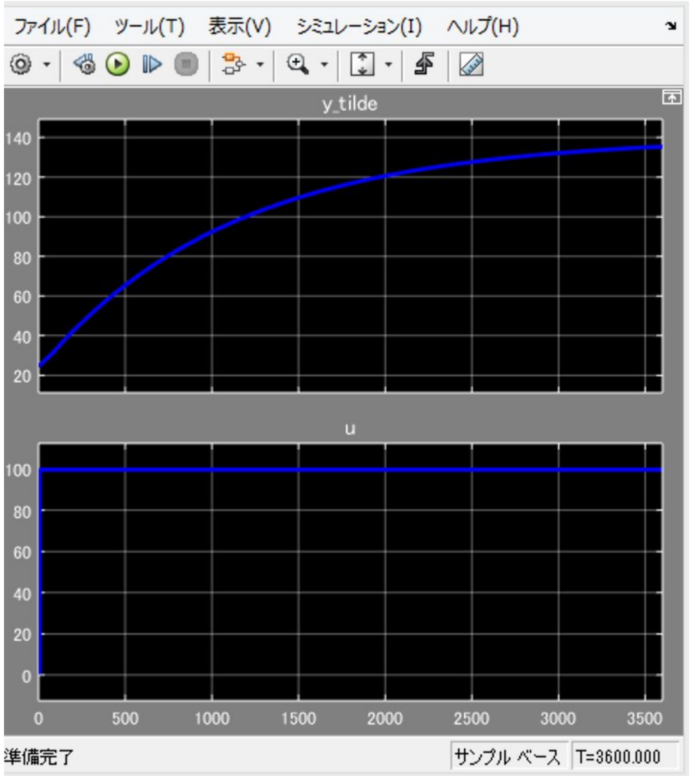
Caution! Do not use large input for step test

Heater Input: 100W

Start: 25℃

Final: 135℃

What's the problem?



Caution! Do not use large input for step test

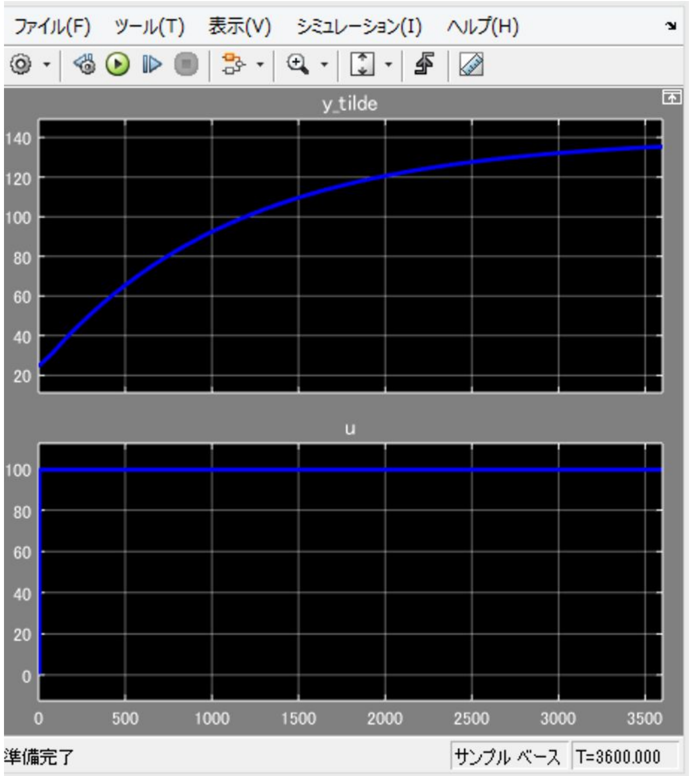
Heater Input: 100W

Start: 25℃

Final: 135℃

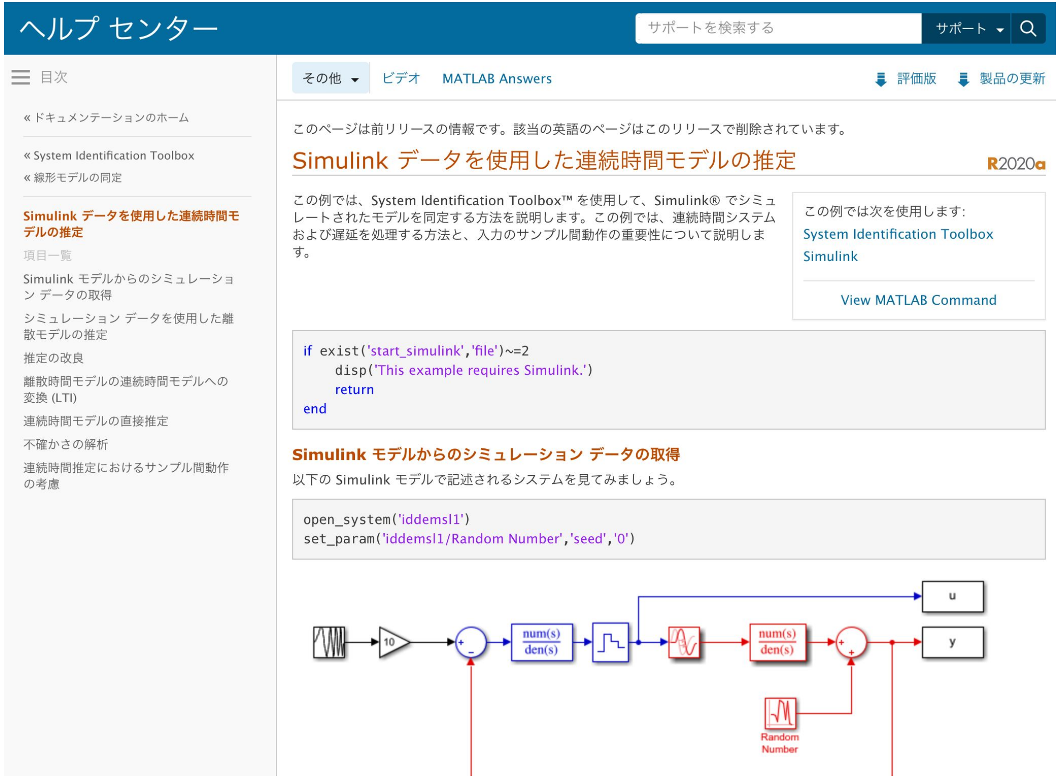
This model does not include **boiling phenomenon.**

We can not measure final temperature as a result.



For advanced study

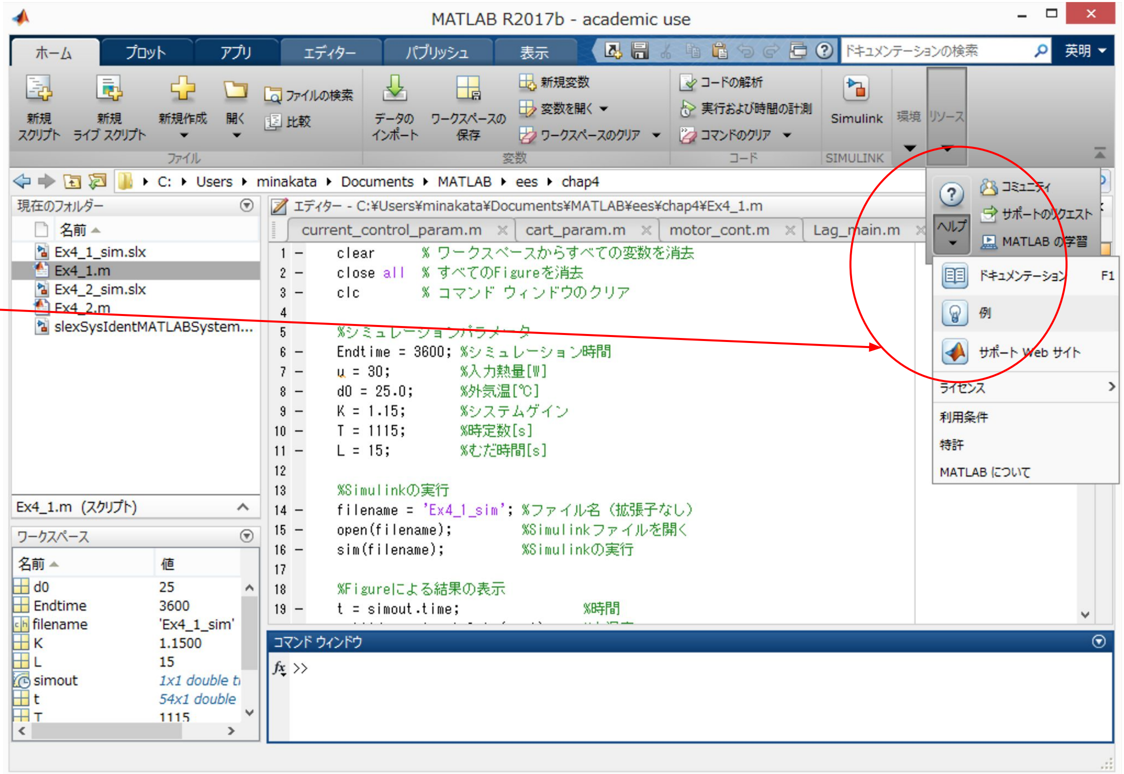
https://jp.mathworks.com/help/ident/examples/estimating-continuous-time-models-using-simulink-data.html



An Example of Digital Filter System Identification

Open from MATLAB Window

help->demo
（ヘルプ->例）



An Example of Digital Filter System Identification

Search "Identification" or "同定"

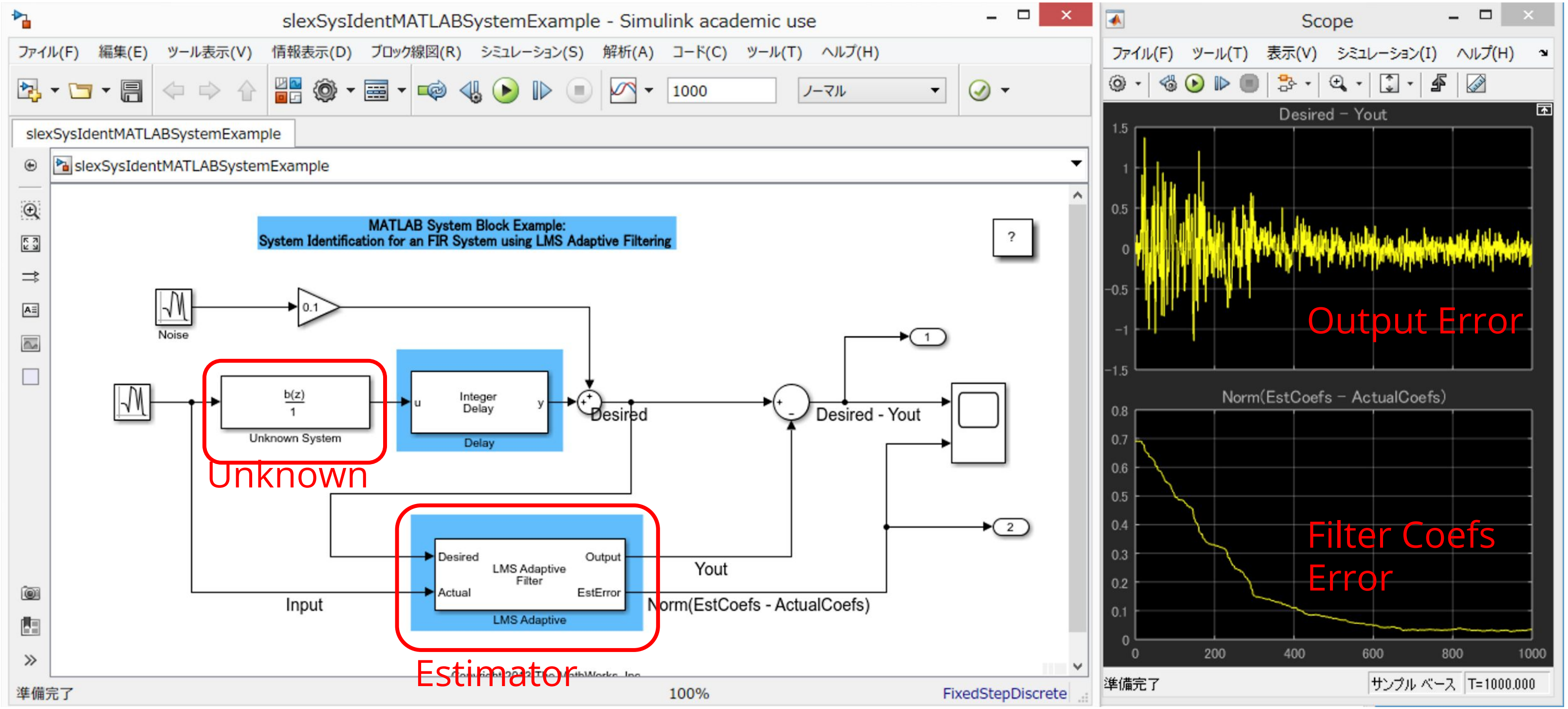
Select "demo" or "例"

Choose this sample.

An Example of Digital Filter System Identification

Open this model.

Output of this sample



Filters Coefs

Unknown

Estimator